

EC-400

GNSS | LTE | BLE

EC-400 is an industrial-grade LTE Cat-1 telematics device built for advanced electric vehicle monitoring, offering deep insights through CAN bus support, wide high-voltage compatibility, Bluetooth connectivity, and IP67-rated rugged protection. Designed for EV fleets, shared mobility, logistics, public transport, and automotive OEMs, it enables reliable real-time tracking, vehicle health visibility, and scalable integration across modern electric mobility ecosystems.



Multiple IO's

9-90V DC Voltage

BLE 5.0 Powered

**Internal High Gain
LTE & GNSS Antenna**

IP67 Casing

CAN Enabled



Electric L3 & L5
Fleet



Renewable Energy
Source Monitoring



AC Monitoring



Logistics and
Supply Chain

INTERFACE

DI, DO, AI	2, 3, 1
CAN, 1-Wire	1, 1
GNSS & LTE Antenna	Internal High Gain Antennas
Indicators	4 LEDs (Power, Charging, LTE, GPS)
USB	2.0 Micro USB
SIM	Nano SIM
Memory	28000+ Log Packets
CAN Protocol Support	All legislated OBD-II Protocols, Non-Legislated OBD Protocols & Heavy-Duty
RS232/ RS485	1*/ 1*

NAVIGATION

GNSS Constellations	GPS, GLONASS, BeiDou, QZSS
Channels	47 Tracking Channels
Horizontal Position Autonomous	1.5m CEP
SBAS	Supported
Tracking Sensitivity	-166 dBm
Re-acquisition Sensitivity	-160 dBm
Velocity	0.1m/s
Hot Start, Warm Start, Cold Start	1 sec, 2 sec, 12 sec

CELLULAR NETWORK

Technology	LTE CAT1, GSM
2G Band	900/1800 MHz
4G LTE Band	LTE FDD: B1/B3/B5/B7/B8/B20
RF Power	GSM/GPRS power level: EGSM900: 4 (33dBm±2dB), DCS1800: 1 (30dBm±2dB); EDGE power level: EGSM900: E2 (27dBm±3dB), DCS1800 : E1 (26dBm+3dB/-4dB); LTE power level: 3 (23dBm±2.7dB)
SMS	Point to point MO & MT, SMS cell broadcast

POWER

Voltage Range	9-90V DC
Internal Battery	1000mAh Li-Po 3.7V

BLUETOOTH

Specification	BLE 5.0
Supported Peripherals	Universal BLE sensors support, Beacons

OPERATING ENVIRONMENT

Working Current	Active Mode < 160mA @ 12V
Operating Temperature	-20 °C to +60 °C (With Battery), -30 °C to +75 °C (Without Battery)
Operating Humidity	5% to 95% non-condensing at +40 °C
Ingress Protection Rating	IP67
Battery charge temperature	0 °C to +45 °C
Battery discharge temperature	-20 °C to +60 °C
Battery storage temperature	-20 °C to +35 °C (For 1 year)

PHYSICAL SPECIFICATION

Dimensions & Weight	94 x 74 x 33 mm (L x B x H) & 143g
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FEATURES

Sensors	Accelerometer, fuel sensor, pressure sensor , all BLE sensors
Configuration and firmware update	FOTA Web, FOTA, Configuration Tool(USB, BLE)
SMS	Configuration, DOUT control, Debug
GPRS commands	Configuration, DOUT control, Debug
Time Synchronization	GPS
Fuel Sensors	LLS (Analog/BLE/RS232), Built-in CAN

*represents optional